

Data Analytics (DA)

Module -4

Python Assignment 2- Data Structures - List, Dictionary, Set & Conditional Statements

Assignment Task Overview

Problem Statement:

As a Python learner, this assignment focuses on working with core data structures such as **lists**, **dictionaries**, and **sets**, along with implementing **conditional statements**.

The objective is to develop a strong understanding of:

- List creation, modification, and access operations
- Dictionary creation and manipulation using key-value pairs
- Set properties and operations like uniqueness, union, and intersection
- Decision-making using conditional statements (**if**, **elif**, **else**)

You are required to write Python programs that demonstrate these concepts and display outputs as per the given tasks.

Tasks:

List (Creation, Modification, and Access):

1. List Creation:

- a. Create a list named `age_list` with five integer elements. For eg., [24, 25, 27, 28 ,29]
- b. Create a list named `name_list` with five string elements.

2. List Operations / Modifications:

- a. Append the string `"Yazhini"` to `name_list`.
- b. Insert the element 26 at index 2 in `age_list`.
- c. Remove the string `"Yazhini"` from `name_list`.
- d. Pop the last element from `age_list`.
- e. Extend the `age_list` with additional ages [31,30,32].
- f. Sort `age_list` in descending order.
- g. Find the max age, min age, and sum of all ages from `age_list`.

3. Accessing List Elements:

- a. Print the first element of `name_list`.
- b. Print the last element of `name_list`.
- c. Print the elements from index 2 to index 4 in `name_list`.
- d. Print the elements of `name_list` in reverse order.

Dictionary (Creation, Modification and Access):

- a. Create a dictionary named `student_marks` that maps the names of five students to their marks (use a scale of 0 to 100).
- b. Access and print the mark of a specific student of your choice.
- c. Add a new student, "Janani," with a mark of 80 to the `student_marks` dictionary.
- d. Update the mark of any one older student to 82.
- e. Use the `keys()`, `values()`, and `items()` methods to print all keys, values, and key-value pairs in the `student_marks` dictionary.

Sets (Operations):

- a. Create a set called `my_set` with the following values: `['a','e','i','o','u','a','a','i']`
Analyze the output and explain the same.
- b. Attempt to access `my_set[4]`. If the code throws an error, provide an explanation.
- c. Create two sets:
`set1` with values: `{1, 3, 5, 7, 9}`
`set2` with values: `{2, 3, 5, 8, 10}`
- d. Compute and print the union and intersection of `set1` and `set2`.

Operators & Conditional Statements : (IF, ELIF, ELSE)

Performance Category Program:

1. Prompt the user for input. Score range should be from 0 to 10 (both inclusive).
2. Find the performance category based on the input score using the following criteria:
 - a. **Above Average:** Score greater than 7
 - b. **Average:** Score between 4 and 7 (both inclusive)
 - c. **Below Average:** Score less than 4
3. **Output:** Print the Performance category
4. **Additional Step:** You can give a prompt of your choice to each category. For example, if the score is below average, "Need to improve your performance; consistent practice will lead to better results".

Sample Output:

```
Enter your score (0 to 10): 7
```

```
Average: Good effort! Keep practicing, there's room for improvement.
```

Deliverables

Submit a project folder containing:

- Complete the assignment in Google Colab or a Jupyter Notebook.
- For Google Colab, share the notebook link with "View" access. If using Jupyter, upload the notebook to Google Drive and ensure "View" access is enabled.
- Put both the Google Colab notebook in a Google Drive folder and share that folder with view access

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
List Operations	Data Structures	Creation, modification, sorting, and analysis of lists	Beginner
List Access	Data Handling	Indexing, slicing, and reversing lists	Beginner
Dictionary Operations	Data Structures	Creating, accessing, updating key-value pairs	Beginner
Dictionary Methods	Built-in Functions	Using <code>keys()</code> , <code>values()</code> , <code>items()</code>	Beginner
Set Operations	Data Structures	Understanding uniqueness, union, intersection	Beginner
Set Limitations	Data Integrity	Handling immutability and indexing errors	Intermediate
Conditional Statements	Control Flow	Using <code>if</code> , <code>elif</code> , <code>else</code> for decision-making	Beginner
Input Handling	Data Handling	Accepting and validating user input	Beginner

DA Skill Taxonomy Structure

Skill / Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial Marks Range
List Operations	3	Correct creation, modification, sorting, and analysis of lists	Minor errors in list operations or calculations	1 – 2
List Access	2	Accurate indexing, slicing, and reversing of list elements	Some access operations incorrect or missing	1
Dictionary Operations	2	Proper creation, access, update, and use of dictionary methods	Minor mistakes in updating or retrieving data	1
Set Operations	2	Correct implementation of uniqueness, union, and intersection with explanation	Partial understanding or missing explanation	1
Conditional Statements	1	Correct use of <code>if</code> , <code>elif</code> , <code>else</code> with proper output	Logic errors or incomplete conditions	0.5

DA Skill Taxonomy Structure

- **List Operations:** Create, modify, sort, and analyze lists
- **Dictionary Operations:** Create, access, update, and retrieve key-value pairs
- **Set Operations:** Understand uniqueness and perform union & intersection
- **Conditional Statements:** Apply `if`, `elif`, `else` for decision-making
- **Input/Output Handling:** Accept user input and display formatted results

Evaluation Guidelines

Review the submitted **.py script** to ensure:

- Correct implementation of list, dictionary, set, and conditional operations
- Code runs without errors
- Proper syntax, indentation, and structure

For each section:

- Map work to defined skills
- Assign marks as per the rubric.

Marking Criteria:

- **Full marks:** Correct logic, accurate output, clean code
- **Partial marks:** Minor errors or incomplete implementation

Additional Checks:

- Outputs are correct
- Code is readable and well-commented
- Feedback includes strengths and improvements

Achievement Criteria & Passing Rules

- Minimum marks required in all sections
- Must demonstrate at least two key skills:
 - List/Dictionary Operations
 - Conditional Statements
- Programs must be executable and logically correct
- Proper input/output handling required
- Provide a **scorecard with feedback** for each submission